

Luca Parolari

PhD Researcher in Computer Vision and Machine Learning

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Summary

PhD researcher in computer vision and machine learning. My research focuses on learning and evaluating robust representations in low-label and noisy settings using weakly supervised, self-supervised, and contrastive learning. I have worked on multimodal vision-language learning, generative models, and benchmarking of text-to-image systems. My profile combines academic research with strong engineering skills in PyTorch, software development, and data-intensive pipeline design, strengthened by previous industry experience.

Selected Research Highlights

- Developed weakly/self-supervised methods for learning visual representations from noisy and low-label data, such as medical videos, using contrastive learning and temporal information.
- Designed models, losses, and evaluation protocols for visual representation learning, with applications to classification, clustering, retrieval, re-identification, and vision-language grounding.
- Built PyTorch/Lightning codebases for data curation, model training, experiment tracking, and reproducible evaluation on multi-GPU/SLURM infrastructure.
- Developed synthetic data generation pipelines for vision-language pre-training and benchmarks for evaluating text-to-image diffusion models.
- Led research projects from problem formulation to implementation, experimental analysis and scientific writing.

Core Skills

AI/ML	Self-supervised & contrastive learning, representation learning, transformers, multimodal vision-language models, diffusion & generative models, model evaluation, data curation
Engineering	Python, PyTorch, PT Lightning, Hugging Face, NumPy, Pandas, scikit-learn, OpenCV, torchvision
Tooling	Multi-GPU distributed training, SLURM, PyTorch DDP, Wandb
Infrastructure	Docker, Docker Compose, Linux, Git, CI/CD, code review
Software	C/C++, Bash, TypeScript/JavaScript, React, React Native, Flask, Django, L ^A T _E X

Research Experience

- 2023–present **PhD Researcher in Computer Vision and Machine Learning**, *University of Padova*, Padova, Italy
- Designed self-supervised and contrastive representation learning methods for complex, noisy, weakly annotated visual data, applied to large-scale endoscopic video understanding.
 - Investigated synthetic data generation for pre-training multimodal models and built large-scale benchmarks to assess text-to-image generation capabilities across semantic and spatial axes.
 - Published work at MICCAI, ISBI, ICIAP, ICPR, and CVPR workshops; received an honorable mention at ICIAP 2025.
- 10/2025– **Visiting PhD Student**, *Telecom Paris – Institut Polytechnique de Paris*, Paris, France
- 03/2026 Worked with the LTCI Lab on self-supervised learning for colonoscopy video understanding, leading to a MICCAI 2026 paper on noisy temporal self-supervision.
- 12/2022– **Research Fellow**, *University of Padova*, Padova, Italy
- 10/2023 Worked on multimodal models for weakly supervised visual grounding, contributing to a BMVC 2023 paper.

Selected Publications

- 2026 **L. Parolari**, P. Gori, L. Ballan, C. Biffi, L. Le Folgoc. *Contrastive Learning under Noisy Temporal Self-Supervision for Colonoscopy Videos*. Accepted at MICCAI 2026.
- 2026 **L. Parolari**, N. Faccioli, L. Ballan. *Benchmarking Layout-Guided Diffusion Models through Unified Semantic-Spatial Evaluation in Closed and Open Settings*. Accepted at CVPRF 2026.
- 2025 **L. Parolari**, A. Cherubini, L. Ballan, C. Biffi. *Temporally-Aware Supervised Contrastive Learning for Polyp Counting in Colonoscopy*. MICCAI 2025.
- 2025 E. Izzo*, **L. Parolari***, D. Vezzarro*, L. Ballan. *7Bench: A Comprehensive Benchmark for Layout-Guided Text-to-Image Models*. ICIAP 2025, honorable mention.

- 2025 **L. Parolari**, A. Cherubini, L. Ballan, C. Biffi. *Towards Polyp Counting in Full-Procedure Colonoscopy Videos*. ISBI 2025.
- 2024 **L. Parolari**, E. Izzo, L. Ballan. *Harlequin: Color-Driven Generation of Synthetic Data for Referring Expression Comprehension*. ICPR 2024.
- 2023 D. Rigoni, **L. Parolari**, L. Serafini, A. Sperduti, L. Ballan. *Weakly-Supervised Visual-Textual Grounding with Semantic Prior Refinement*. BMVC 2023.

Industry Experience

- 01/2022–
11/2022 **AI Software Engineer**, *Orobix*, Bergamo, Italy
Developed AI products in computer vision and optimization planning, contributing across the full ML lifecycle from requirements gathering to deployment and maintenance. Built production-oriented AI and software infrastructures for cloud and edge environments, using Python, Django, React, and containerized services while collaborating with technical and business stakeholders.
- 2016–2021 **Software Engineer / Full Stack Developer**, *Freelance*, Italy
Designed and delivered web, mobile, and embedded software for industrial operations, asset management, fiscal reporting, and RTU/radio-system interfaces. Worked with Python, PHP/Laravel, Vue.js, Java, Android, React Native, Raspberry Pi, and Arduino.

Education

- 2023–present **PhD in Computer Vision and Machine Learning**, *University of Padova*, Padova, Italy
- 2019–2021 **Master's Degree, Computer Science and Artificial Intelligence**, *University of Padova*, Padova, Italy
110/110 cum laude. Thesis: *Weakly Supervised Visual-Textual Grounding based on Concept Similarity*.
- 2016–2019 **Bachelor's Degree, Computer Science**, *University of Parma*, Parma, Italy
110/110 cum laude.

Teaching, Mentorship and Service

- Teaching Teaching Assistant for Machine Learning and Programming courses at the University of Padova; delivered lectures/labs, prepared assignments, supported student projects, and assisted with grading.
- Mentorship Co-supervised Bachelor's and Master's students in Computer Science and Data Science, supporting project design, experimental design, and scientific writing.
- Reviewing Reviewer for international journals and conferences in computer vision, machine learning, and medical image analysis.

Languages and Interests

- Languages Italian: native. English: advanced/professional (C1). French: elementary (A1).
- Interests Ski mountaineering, cycling, astronomy.